

5.36 Solution: (a) From $\mathbf{E} = -\partial\mathbf{A}/\partial t$ and also the vector potential from Problem 5.35, the electric field is

$$E_\phi = \frac{3B_0a}{\pi} \sin\theta \int_0^{+\infty} \nu k^2 e^{-\nu t k^2} j_1(k) j_1\left(\frac{kr}{a}\right) dk.$$

In the limit $t \rightarrow 0^+$, the electric field becomes

$$\begin{aligned} E_\phi &= \frac{3B_0a\nu}{\pi} \sin\theta \int_0^{+\infty} k^2 j_1(k) j_1\left(\frac{kr}{a}\right) dk \\ &= \frac{3B_0a\nu}{\pi} \sin\theta \cdot \frac{\pi}{2} \delta\left(1 - \frac{r}{a}\right) \\ &= \frac{3B_0a}{2\mu\sigma a^2} \sin\theta \cdot a\delta(r - a) \\ &= \frac{3B_0}{2\mu\sigma} \sin\theta \delta(r - a), \end{aligned}$$

and the current density is

$$\mathbf{J} = \sigma\mathbf{E} = \frac{3B_0}{2\mu} \sin\theta \delta(r - a) \hat{\phi}.$$

This agrees with Problem 5.35 (a).

For $\nu t \gg 1$, applying the same procedure as in Problem 5.35 (c), we have

$$\begin{aligned} E_\phi &= \frac{3B_0a}{\pi} \sin\theta \frac{\nu}{(\nu t)^{3/2}} \int_0^{+\infty} e^{-u^2} u^2 j_1\left(\frac{u}{\sqrt{\nu t}}\right) j_1\left(\frac{u}{\sqrt{\nu t}} \frac{r}{a}\right) du \\ &\approx \frac{3B_0a}{\pi} \sin\theta \frac{\nu}{(\nu t)^{3/2}} \int_0^{+\infty} e^{-u^2} u^2 \frac{u}{3\sqrt{\nu t}} \frac{u}{3\sqrt{\nu t}} \frac{r}{a} du \\ &= \frac{3B_0r}{\pi} \sin\theta \frac{\nu}{(\nu t)^{5/2}} \frac{3\sqrt{\pi}}{72} \\ &= \frac{B_0\nu}{8\sqrt{\pi}(\nu t)^{5/2}} r \sin\theta. \end{aligned}$$

(b) The power dissipated is given by

$$\begin{aligned} P &= \int \mathbf{J} \cdot \mathbf{E} d^3x = \sigma \int |\mathbf{E}|^2 d^3x \\ &= \frac{9\sigma B_0^2 a^2}{\pi^2} \int d^3x \sin^2\theta \int_0^{+\infty} \nu k_1^2 e^{-\nu t k_1^2} j_1(k_1) j_1\left(\frac{k_1 r}{a}\right) dk_1 \\ &\quad \cdot \int_0^{+\infty} \nu k_2^2 e^{-\nu t k_2^2} j_1(k_2) j_1\left(\frac{k_2 r}{a}\right) dk_2 \\ &= \frac{9\sigma B_0^2 a^2}{\pi^2} \nu^2 \int_0^{+\infty} dk_1 \int_0^{+\infty} dk_2 k_1^2 e^{-\nu t k_1^2} j_1(k_1) \cdot k_2^2 e^{-\nu t k_2^2} j_1(k_2) \\ &\quad \cdot \int d\Omega \sin^2\theta \int_0^{+\infty} r^2 j_1\left(\frac{k_2 r}{a}\right) j_1\left(\frac{k_1 r}{a}\right) dr \\ &= \frac{9\sigma B_0^2 a^2}{\pi^2} \nu^2 \cdot \frac{8\pi}{3} \int_0^{+\infty} dk_1 \int_0^{+\infty} dk_2 k_1^2 e^{-\nu t k_1^2} j_1(k_1) \cdot k_2^2 e^{-\nu t k_2^2} j_1(k_2) \cdot \frac{\pi}{2(k_1/a)^2} \delta\left(\frac{k_1}{a} - \frac{k_2}{a}\right) \\ &= \frac{9\sigma B_0^2 a^2}{\pi^2} \nu^2 \cdot \frac{8\pi}{3} \cdot \frac{\pi}{2} a^3 \int_0^{+\infty} e^{-2\nu t k^2} [k j_1(k)]^2 dk \end{aligned}$$

$$= \frac{12B_0^2 a^3 \nu}{\mu} \int_0^{+\infty} e^{-2\nu t k^2} [kj_1(k)]^2 dk.$$

Clearly,

$$P = -\frac{dW_m}{dt}.$$

(c) The power dissipated can also be expressed as

$$P = 12B_0^2 a^5 \nu^2 \sigma \int_0^{+\infty} e^{-2\nu t k^2} [kj_1(k)]^2 dk,$$

since $\nu = (\mu\sigma a^2)^{-1}$. Then, the total electric energy is

$$W_e = \frac{\epsilon_0}{2} \int |\mathbf{E}|^2 d^3x = \frac{\epsilon_0}{2\sigma} P = 6\epsilon_0 B_0^2 a^5 \nu^2 \int_0^{+\infty} e^{-2\nu t k^2} [kj_1(k)]^2 dk.$$

If we include the displacement current effect in Eq. (5.160), it will become

$$\nabla^2 \mathbf{A} = \mu\sigma \frac{\partial \mathbf{A}}{\partial t} + \mu\epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2},$$

and there will be an extra contribution to the magnetic energy,

$$W_m^{(1)} = -\frac{1}{2\mu} \int \mathbf{A} \cdot \mu\epsilon_0 \frac{\partial^2 \mathbf{A}}{\partial t^2} d^3x.$$

By repeating the same calculation as in Problem 5.35 (c), we can get

$$W_m^{(1)} = -6\epsilon_0 B_0^2 a^5 \nu^2 \int_0^{+\infty} e^{-2\nu t k^2} [kj_1(k)]^2 dk,$$

which is identical to the electric energy, but only differs in sign. Therefore, the total electromagnetic energy is given by the magnetic energy found in Problem 5.35 (c), and power dissipated found in part (b) correctly accounts for the time derivative of the electromagnetic energy.

Of course, we have only treated the displacement current as a perturbation, which needs to be justified. We can see that, in the limit of $\nu t \gg 1$, W_e and $W_m^{(1)}$ behave asymptotically as $(\nu t)^{-5/2}$. Compared to the asymptotic behavior of W_m as $(\nu t)^{-3/2}$, the electric energy and the magnetic energy due to displacement current are indeed small and can be neglected.